TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	78.5 ± 1.43
Correlation Stability	88.1 ± 0.94
Distribution Consistency	87.7 ± 0.46
Entropy Preservation	59.1 ± 1.37
Mean Z Score Drift	0.4 ± 0.02
Categorical Fidelity	81.6 ± 2.87
Privacy Risk	38.2 ± 3.49
Data Density	100.0 ± 0.00
Coverage Score	16.0 ± 3.85
Feature Accuracy	63.8 ± 0.87
Novelty Score	0.7 ± 0.02

TabuLLM: AI Benchmark Report

Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	41.012	-3.735	8.3%	OK
TB	3.299	1.219	-2.080	63.0%	Drift
DB	1.486	0.438	-1.048	70.5%	Drift
Alkphos	290.576	231.267	-59.309	20.4%	Warn
Sgpt	80.714	33.448	-47.266	58.6%	Drift
Sgot	109.911	41.023	-68.888	62.7%	Drift
TP	6.483	6.425	-0.058	0.9%	OK
ALB	3.142	3.232	+0.090	2.9%	OK
A/G Ratio	0.947	0.992	+0.045	4.8%	OK
Selector	1.286	2.000	+0.714	55.5%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	78.5 ± 1.43
Correlation Stability (*2)	88.1 ± 0.94
Distribution Consistency (*3)	87.7 ± 0.46
Entropy Preservation (*4)	59.1 ± 1.37
Mean Z Score Drift (*5)	0.4 ± 0.02
Categorical Fidelity (*6)	81.6 ± 2.87
Privacy Risk (*7)	38.2 ± 3.49
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	16.0 ± 3.85

Feature Accuracy (*10)	63.8 ± 0.87
Novelty Score (*11)	0.7 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6829059829059829	N/A	0.6572649572649573	-0.02564102564102565	
F1	0.7891024130608028	N/A	0.7476605021996151	-0.04144191086118759	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	N/A	0.6923076923076923	-0.010256410256410243	
F1	0.7997903571549948	N/A	0.7822348569197645	-0.017555500235230705	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6615384615384616	N/A	0.6111111111111111	-0.05042735042735046	*
F1	0.7653043027384351	N/A	0.6965626887658671	-0.06874161397256791	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743589	N/A	0.5572649572649573	-0.0017094017094017033	
F1	0.5587432119380917	N/A	0.5541493497842884	-0.004593862153803285	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	N/A	0.5675213675213675	-0.13931623931623932	*
F1	0.8279091624529537	N/A	0.5915347072672636	-0.23637445518569003	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	N/A	0.6282051282051282	-0.04615384615384614	*
F1	0.7821343746683586	N/A	0.7023122529398937	-0.07982212172846484	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7128205128205128	N/A	0.6538461538461538	-0.05897435897435895	*
F1	0.8148680446178964	N/A	0.7302258817184218	-0.08464216289947453	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	N/A	0.6384615384615384	-0.06752136752136753	*
F1	0.8128002697439015	N/A	0.6969904478934851	-0.11580982185041636	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7136752136752136	N/A	0.6418803418803419	-0.07179487179487178	*
F1	0.8105980826302283	N/A	0.6989714313988609	-0.11162665123136721	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	N/A	0.6350427350427351	-0.05128205128205129	*
F1	0.7971299319768493	N/A	0.6875045705426326	-0.10962536143421682	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	41	Female	0.9	0.2	167	23	21	6.81765834536006	3.41863403479337	0.92990243105666
								9	4	96
2	45	Female	1.09315723263333	0.43452578610666	156	16	17	7.42084025137332	3.83452578610666	1.08273710694666
			19	55				9	53	74
3	36	Female	1.36905303872210	0.49660173641263	364	147	86	7.97396913960737	4.23094696127789	1.06377434884526
			25					5	8	38
4	34	Female	1.16378187338906	0.38189093669453	355	152	87	8.23360010121329	4.35170916451875	1.09396364556484
			77	394				1	7	47
5	16	Male	1.19048466950431	0.59048466950431	314	13	19	6.57495548510237	2.67495548510237	0.63689416311961
								2	22	24
6	28	Female	0.62836470470595	0.10945490156865	178	36	29	6.93781960627461	4.15672940941191	1.43781960627461
			8	265				1	5	05
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	31	Male	0.85182935461456	0.12408532269271	469	25	34	6.21310919500532	3.11310919500532	0.99634129077086
			52	742				7	62	97
2	29	Male	1.22991608986502	0.38621853369938	207	53	130	6.71378146630061	3.69882342136809	1.57697464328527
			94	597				4	95	78
3	35	Male	1.73919217930430	0.55946145286953	279	50	162	6.39865363217383	3.15946145286953	0.92026927356523
			04	35				4	37	32
4	27	Male	1.01383826314052	0.30518934867769	181	55	112	6.79481065132230	3.88789151975204	1.83356706252063
			16	554				46	4	08
5	35	Male	1.74411365366800	0.90808524025100	230	37	125	6.29853407072066	3.04118179510932	0.92367840072713
			5	4				05	67	08
6	38	Male	0.80470645308414	0.20352983981311	246	55	91	7.40352983981311	4.29823508009344	1.37776443478502
			97	22				2	4	89
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

TabuLLM: AI Benchmark Report

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	31	Male	0.8767522967252807	0.2589174322417602	213	18	23	7.3	4.2767522967252801	1.3767522967252805
2	25	Male	0.8971167357208181	0.2985583678604069	159	24	25	6.8884669428832703	4.3798171500457207	1.6893319221670269
3	35	Male	0.7603783213973505	0.2	185	18	17	5.8415132855894002	3.1811349641920517	1.0817789372177278
4	31	Male	1.0590831965725864	0.4118845522642707	180	15	16	6.2594227613213503	3.1943972886166314	0.9650584328730306
5	54	Male	0.8966912334659232	0.2	116	35	16	6.2330876653407685	3.2397051984089225	1.0165438326703842
6	63	Male	1.0813799082316042	0.4720698623474617	142	21	23	5.5651703211893855	3.2	1.3536908210516279
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	32	Female	0.7901545534849231	0.1901545534849231	135	30	28	7.2393817860603007	4.3901545534849244	1.4803091069698462
2	35	Female	0.8740592994421891	0.2	193	37	35	7.7928733105984008	4.6740592994421899	1.5924743955375122
3	40	Female	0.8900672134174066	0.2	201	31	25	7.7589245853214955	3.8794622926607474	1.0
4	49	Female	0.6396846084867883	0.1396846084867838	187	17	21	6.9174768678943007	3.4158999103282499	0.9381076509207302
5	52	Female	0.6	0.1	185	17	24	6.4804977029345645	3.0195022970654346	0.8195022970654349
6	28	Female	1.1396556988116044	0.3597704658744027	181	58	38	6.3413772047535835	3.5816067388791812	1.2408033694395908
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

TabuLLM: AI Benchmark Report

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	30	Male	0.7181074566657469	0.2	187	23	45	7.291355260005519	3.1724298266629898	0.7543223699972409
2	26	Male	0.7508948555477339	0.2	185	27	50	7.094630866713596	3.3035794221909356	0.8526845666432018
3	32	Male	0.8	0.2	182	31	39	6.491016693597731	3.7317374798017013	1.308983306402269
4	37	Female	0.7992471151175373	0.2	145	19	23	6.098494230235074	3.1	1.0305270194177238
5	43	Male	4.4635640981195275	2.0488241161591167	1095	60	50	5.732957932900647	2.5215965548344102	0.7329579329006471
6	49	Male	7.2573280937110002	3.5658624749688017	1498	82	62	5.642671906288998	2.3	0.6
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	48	Female	0.8	0.2	234	10	14	6.614364028456494	2.957090647695671	0.8227997124652338
2	39	Male	0.9767550101272782	0.28837750506363913	218	19	22	7.230265030381835	4.225662575954587	1.4139935374709294
3	31	Male	0.8260346936185371	0.24201156453951236	231	21	23	6.140231290790247	3.18224285532976	1.0360925163160909
4	27	Male	0.7035994254855498	0.2	242	21	23	5.296400574514450 5	2.307198850971099 4	0.7071988509710996
5	42	Male	0.8836426743605813	0.2	167	21	17	6.181786628197093	3.098143953836512 3	0.9490719769182562
6	40	Male	0.9184524308087916	0.20790818463233923	168	21	18	6.131632738529356 5	3.013180307720565	0.9005272123088226
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

TabuLLM: AI Benchmark Report

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	39	Male	1.11161647559618 08	0.40774431706412 06	184	50	66	7.14214299596361 8	4.02300757247839	1.26924810568804 03
2	60	Male	1.54600057641031 77	0.64600057641031 78	194	60	84	6.51599769435872 9	3.73099913538452 33	1.32300028820515 88
3	26	Male	1.0	0.3	172	52	93	6.93097506606069 7	3.81268328929286 86	1.47890397949469 17
4	30	Male	0.95790962032744 02	0.27895481016372 01	177	54	110	6.60007069655534 1	3.76320626606418 07	1.77108053811395 05
5	38	Male	2.24173896601276 96	0.90834779320255 39	172	71	107	5.68330441359489 2	3.28330441359489 23	1.29165220679744 6
6	25	Male	2.60365899628795 73	0.98073179925759 15	162	79	122	5.53853640148481 75	3.13853640148481 7	1.21926820074240 85
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	19	Male	1.27464794839096	0.67464794839096 02	178	14	24	7.67408466581649 7	4.37436630710372 75	1.27492958967819 2
2	62	Male	0.87322405923204 79	0.2	191	51	45	6.17404361499168 9	3.92677594076795 2	1.76297819250415 57
3	61	Male	1.00554185590629 16	0.30554185590629 157	191	50	47	6.07538703993306 6	3.9	1.79722907204685 42
4	37	Male	0.61219393184448 31	0.10609696592224 155	164	22	34	5.90609696592224 15	2.85732123854430 9	0.87561213631103 39
5	5	Male	0.60856413399784 39	0.10285471133261 465	290	37	30	4.89135076264366 8	2.05709422665229 27	0.70856413399784 39
6	5	Male	0.9	0.21425565887625 37	342	28	31	8.01425565887625 3	4.02851131775250 8	1.0
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Female	0.7520445592118081	0.1	145	21	26	6.427509848670111	3.5836435263055346	1.2375465289458303
2	50	Female	0.7524107529481125	0.15241075294811252	142	24	21	6.633124729363212	3.5807139764151	1.1427677411556625
3	32	Male	3.769400468397126	1.5880932411149424	151	37	84	5.811906758885057	2.991548717386782	1.0
4	47	Female	0.8903418112256911	0.2	174	23	53	5.451709056128455	2.637221772966992	0.8782690752578048
5	37	Female	0.7872791789080653	0.2	158	27	40	5.80918768362098	2.0982334312645228	0.4936395894540326 3
6	19	Female	0.9050818776000206	0.26836062586667353	142	24	25	6.653114993066612 5	3.626557496533306	1.1936721251733347
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	52	Female	1.0114564309359941	0.3762136924466618	158	15	13	6.00087344568135	3.212329876617344	1.1295145230213353
2	41	Male	0.8129010482948172	0.2752673655298782	201	67	38	6.07598034027622	2.600712974746342	0.7376336827649 39
3	44	Male	1.8346163775065838	0.6826918112467081	217	64	49	6.650955829337143	3.4134590562335405	1.0586540943766 46
4	24	Female	0.6925881359652682	0.2	187	11	10	5.477764407895805	2.3296474561389267	0.7276938200084535
5	75	Female	1.0	0.3	205	21	14	6.180958831867682	3.1182110775692255	0.9518211077569225
6	36	Male	0.8	0.2	158	26	22	7.27637889557135	3.772442044833241	1.0401578731547 29
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3